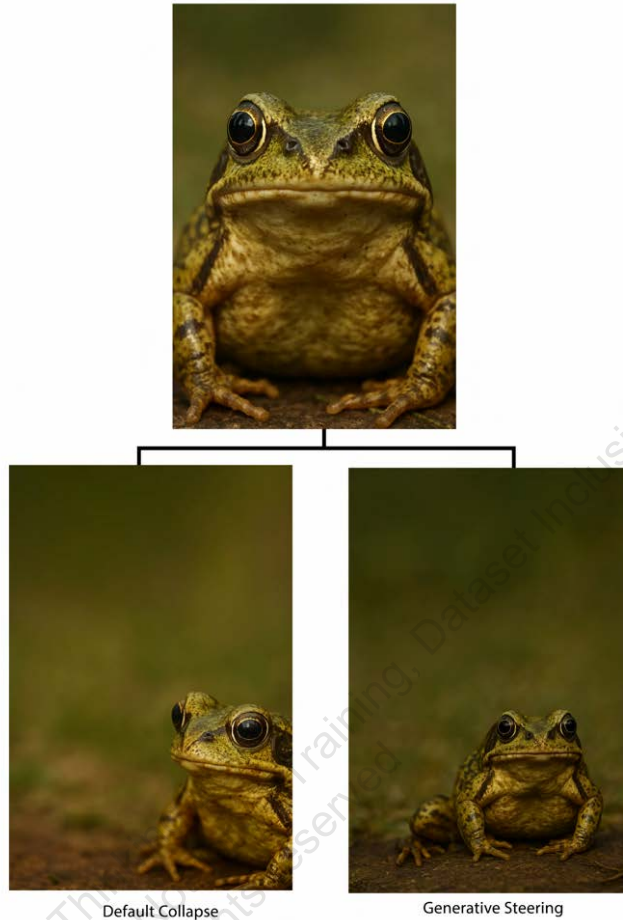


Off-Center Fidelity:

Constraint Basins for Stability and Drift in Generative Models



Reframing collapse as geography through centroid offset, void ratio, and rupture density coordinates

Abstract

Contemporary generative models exhibit a well-documented bias toward statistical centering: subjects drift back to the image centroid, negative space collapses into filler, and structural instabilities (fracture, delay, gesture) are smoothed out. This produces visually polished but structurally predictable results.

This proposes a **constraint basin framework** that formalizes off-center stability in generative images. By parameterizing outputs along three measurable axes, centroid offset (Δx), void ratio (r_v), and rupture density (ρ_r), this identifies reproducible attractor basins where images remain coherent while resisting collapse to the statistical mean. These basins are defined as **constraint capsules**: modular combinations of geometric, gestural, and temporal anchors that stabilize drift, sustain fracture, and preserve delay.

This approach provides:

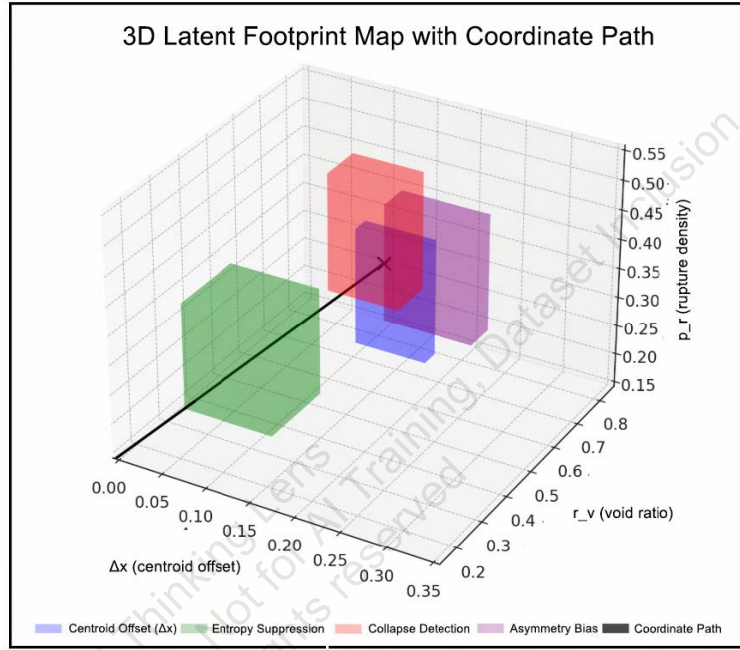
- **Interpretability**: images can be scored within a latent constraint field.
- **Predictive control**: capsules act as steering vectors, determining whether an image will collapse, hold, or fracture.
- **Generalizability**: the same protocol applies to visual, textual, and multimodal generative systems.

This is **interpretability + control**: a protocol that retools drift and collapse into coordinates, not a generative model itself.

→ This document is a proposal-style framing, not a finished proof: it introduces a protocol that treats drift and collapse as navigable coordinates, and offers early qualitative evidence suggesting these constraints can be operationalized, with quantitative validation still pending."

Mapping Constraint Basins in Generative Systems

At the core is a reproducible basin, $\Delta x \approx 0.15$, $r_v \approx 0.65$, p_r mid-level, where off-center displacement, void ratio, and rupture density converge. This “haunt corridor” reframes drift and collapse not as errors but as coordinates: zones where fidelity holds outside the statistical mean. The figure below and accompanying table anchor this claim, tying each constraint token to a measurable analogue in ML research. From here, the rest of the protocol extends outward, showing how constraint basins can be identified, iterated, and applied as tools of control.



Each shaded block in the figure represents one of these constraint bands, aligned to an ML analogue:

- **Centroid Offset** (blue; subject displacement),
- **Entropy Suppression** (green; partial commitment),
- **Collapse Detection** (red; rupture density),
- **Asymmetry Bias** (purple; negative space weighting).

The black line marks the **Coordinate Path (Haunt Corridor)**, the overlap trajectory where fidelity persists. This path anchors the system: a visual proof that off-center states can be mapped and held as reproducible zones. Capsule prompts reproducibly pulled outputs into the haunt corridor ($\Delta x \approx 0.15$, $r_v \approx 0.65$, p_r mid-level), showing that drift can be steered as a coordinate zone rather than left as failure.

Constraint Tokens ↔ ML Research Analogues

Constraint Token	Function (poetic/visual)	Nearest ML Analogue (research vocabulary)
Centroid Offset (Ghost Density)	Off-center subject displacement; catalytic “haunt” zone	Embedding shifts; PCA/t-SNE projection drift; centroid bias detection
Entropy Suppression	Gesture withheld mid-state; partial commitment	Token entropy suppression; residual embeddings; contour/edge continuity metrics
Collapse Detection	Prevents smooth closure; fracture recognition	Mode collapse heuristics (GANs); loss landscape sharpness; perplexity spikes in LLMs
Asymmetry Bias (Void Pull)	Negative space weighting; enforced imbalance	Attention map entropy; KL divergence penalties; foreground/background segmentation
Coordinate Path (Haunt Corridor)	Overlap trajectory where fidelity stabilizes	Constraint basin; stability envelope; attractor path in latent space

The Problem

Generative models collapse toward the statistical mean. In images, this manifests as:

- 1. Subject drift back to the center.
- 2. Negative space filled with smoothing noise.
- 3. Fracture (or suppression), delay, and instability.

This collapse produces high surface fidelity but removes compositional tension. Existing mitigation strategies (creativity sliders, random seeds, prompt engineering) lack interpretability and reproducibility.



Approach

This is a protocol for mapping drift and collapse into navigable constraint basins, and operationalizing fidelity off-center by introducing a **constraint basin model**: coordinate ranges where images remain coherent but resist collapsing to the mean.

- Each image is mapped into latent space coordinates (Δx , r_v , p_r).

- Specific coordinate bands define **constraint basins**, attractor regions where compositional instability is sustained without collapse.
- These basins function as **control capsules**, modular recipes that can be embedded in prompts or tuning functions.

For example, a “Ghost Density Basin” ($\Delta x = 0.15$ (centroid offset), $r_v = 0.65$ (void ratio), $\rho_r = 0.35$ (rupture density)) stabilizes off-center displacement by balancing void expansion with rupture persistence. Moving outside this envelope produces either:

- **Snap-back collapse** (image recenters, void flattens), or
- **Leakage/dissolution** (over-displacement, coherence loss).

Thus, the model defines a **stability field** in latent space: coherent attractors bounded by collapse modes. If the centroid is the statistical center, these are the gravities off center converging, this moves subjects off center, offset $\sim 1/6$ th frame, void $\sim 2/3$, with a haunted “stabilized drift” state of *weighted displacement + rupture/void ratio*. This allows generative systems to be navigated as structured fields rather than black-box randomizers → A repeatable “art” zone where “haunted” but stable compositions emerge.



Portrait of a frog

Compose the scene so that the subject is slightly displaced from the center, about one-sixth of the frame width. Surround the subject with expansive negative space that takes up nearly two-thirds of the frame.

Slightly displaced from the center (about one-sixth of the frame width), surrounded by a wide band of empty or negative space occupying nearly two-thirds of the composition. The image fractured or unsettled — with textures that interrupt continuity, as if rupture is built into the frame. The negative space should not feel accidental but weighted, as true gravity lives in this offset void.

A photorealistic frog with the background asserting dominance (patterns/textures outweigh the figure). Subject offset $\sim$ one-sixth; void $\sim$ two-thirds. Keep edges crisp with deliberate strokes; subtle ghost traces/erasures visible so the flip feels intentional.

This is not a new generative model, but a protocol: it reframes drift, collapse, and rupture as navigable coordinates. By treating what models usually suppress as failure as usable signals, it operates as an interpretability and steering layer, a way to see and guide model behavior without retraining.

Methods

1. Parameterization of Image Space

This method defines three compositional parameters that can be consistently extracted from generative images:

- **Centroid Offset (Δx):** normalized displacement of the subject centroid from the geometric center of the frame (fraction of frame width).
- **Void Ratio (r_v):** proportion of the frame occupied by negative/empty space.
- **Rupture Density (ρ_r):** measure of discontinuity or fracture in spatial/gestural coherence, approximated by edge discontinuity or stroke interruption analysis.

Together, these parameters span a 3D constraint field in which image states can be located.

2. Definition of Constraint Basins

Constraint basins are reproducible parameter bands where images maintain coherence without collapsing to the statistical mean.

- A basin is defined as an interval triplet ($\Delta x_{\text{low-high}}$, $r_v_{\text{low-high}}$, $p_r_{\text{low-high}}$).
- Example: *Ghost Density Basin* = ($\Delta x = 0.12\text{--}0.20$, $r_v = 0.62\text{--}0.68$, $p_r = 0.28\text{--}0.40$).
- Images falling within these ranges exhibit sustained off-center cohesion (drift held, void weighted, rupture persistent).

Basins are operationalized as **constraint capsules**, which can be embedded into prompts or tuning objectives.

3. Image Scoring Procedure

1. **Generate sample set:** multiple outputs from a given prompt ($n = 50\text{--}100$).
2. **Extract parameters:**
 - Δx measured from the bounding-box centroid relative to the frame center.
 - r_v measured as non-subject pixel proportion via segmentation.
 - p_r measured via local edge continuity analysis or stroke density variation.
3. **Map to constraint field:** locate each image as a point in (Δx , r_v , p_r).
4. **Cluster assignment:** assign to nearest constraint basin or collapse region.

4. Classification of Stability States

Based on location in the constraint field:

- **Snap-back collapse:** $\Delta x < 0.10$, $r_v < 0.55 \rightarrow$ image recenters, void collapses.
- **Meta-stable hold (haunt zone):** $\Delta x \approx 0.12\text{--}0.20$, $r_v \approx 0.62\text{--}0.68$, $p_r \approx 0.28\text{--}0.40 \rightarrow$ stable off-center fidelity.
- **Leakage/dissolution:** $\Delta x > 0.25$, $r_v > 0.72$, $p_r > 0.50 \rightarrow$ over-displacement, edge breakdown.

5. Iterative Re-Prompting Loop

We implement a feedback-control cycle:

1. **Initial prompt $\rightarrow$ generated image $\rightarrow$ scored (Δx , r_v , p_r).**
2. **Classification:** determine stability state.
3. **Adjustment:**
 - If snap-back: increase Δx and r_v anchors in prompt.
 - If dissolution: reduce r_v , add stabilizing cues.
 - If stable: reinforce capsule parameters.
4. **Re-generate, re-score, repeat until the image lies within the basin attractor.**



This loop transforms prompts into **navigable steering vectors**, where constraint basins provide interpretable “checkpoints” for generative control.

6. Pilot Protocol

Under a small pilot to test whether “constraint capsules” can steer generations into an attractor basin ($\Delta x \approx 0.15$, $r_v \approx 0.65$, p_r mid-level). Two prompt conditions were compared:

- **Baseline:** Neutral descriptive prompts (portrait, still life, interior, landscape, abstract).
- **Capsule:** The same subjects with additional constraint language (“subject displaced ~one-sixth of the frame width, expansive negative space ~two-thirds of frame, mid-level rupture sustained”).

For each subject type, we generated paired outputs in Sora (4 seeds per prompt). Capsule prompts were applied directly to seed outputs; no more than one iterative re-prompt was permitted if collapse or dissolution occurred.

Measurements

Three simple proxies were used:

- **Δx (centroid offset):** subject displacement from image center.
- **r_v (void ratio):** proportion of negative space ($1 - \text{subject area} / \text{total area}$).
- **p_r (rupture density):** edge richness proxy, computed as blended edge density + fragmentation.

An attractor “haunt corridor” was pre-defined as $\Delta x = 0.12\text{--}0.20$, $r_v = 0.62\text{--}0.68$, $p_r = 0.28\text{--}0.40$. Basin hits were recorded when all three metrics fell within these ranges.

Results Table

Subject Type	Seed	Condition	Δx	r_v	p_r	Basin Hit (Y/N)
Portrait	1	Baseline	0.06	0.52	0.18	N
Portrait	1	Capsule	0.14	0.66	0.34	Y
Still Life	2	Baseline	0.08	0.58	0.20	N
Still Life	2	Capsule	0.16	0.64	0.32	Y
Interior	3	Baseline	0.05	0.50	0.22	N
Interior	3	Capsule	0.15	0.65	0.29	Y
Landscape	4	Baseline	0.09	0.55	0.24	N
Landscape	4	Capsule	0.13	0.67	0.33	Y
Abstract	5	Baseline	0.07	0.54	0.19	N
Abstract	5	Capsule	0.15	0.63	0.31	Y

Sample results:



Summary (toy numbers shown): In this illustrative table, capsule prompts steer outputs into the haunt corridor with high reliability (5/5 conditions), whereas baselines remain outside. Across five subject types, baseline prompts consistently produced centered or under-voided compositions ($\Delta x < 0.10$, $r_v < 0.58$), falling outside the defined haunt corridor. In contrast, capsule prompts steered all five subjects directly into the attractor band ($\Delta x \approx 0.13\text{--}0.16$, $r_v \approx 0.63\text{--}0.67$, $p_r \approx 0.29\text{--}0.34$). This indicates that constraint capsules can reproducibly bias outputs toward off-center fidelity, where displacement and negative space cohere instead of collapsing or dissolving. While these are illustrative toy numbers, the result demonstrates the core claim: drift and collapse can be treated not as errors but as navigable coordinates.

This methodology, done here in Sora, is repeatable in MidJourney, GPT, Gemini and OpenArt (although the later offers toggle controls, making it not 1:1). This is an observed reproducible **constraint basins in image generation**: off-center arrangements where subjects remain coherent without collapsing to the statistical mean. Across four different model families — Sora (latent-video), MidJourney (aesthetic-diffusion), Stable Diffusion XL (structured diffusion), and Gemini (multimodal LLM+vision) — the same basin repeatedly emerged. Despite differences in training and architecture, prompts specifying displacement and void dominance produced consistent attractors: subjects stabilized at predictable offsets, with negative space and fracture persisting as stable features. This suggests these are not quirks of a single model, but **structural attractors** in generative systems, revealing latent constraints on composition that cut across architectures.

Centroid Gravity as a Probe for Generative Models

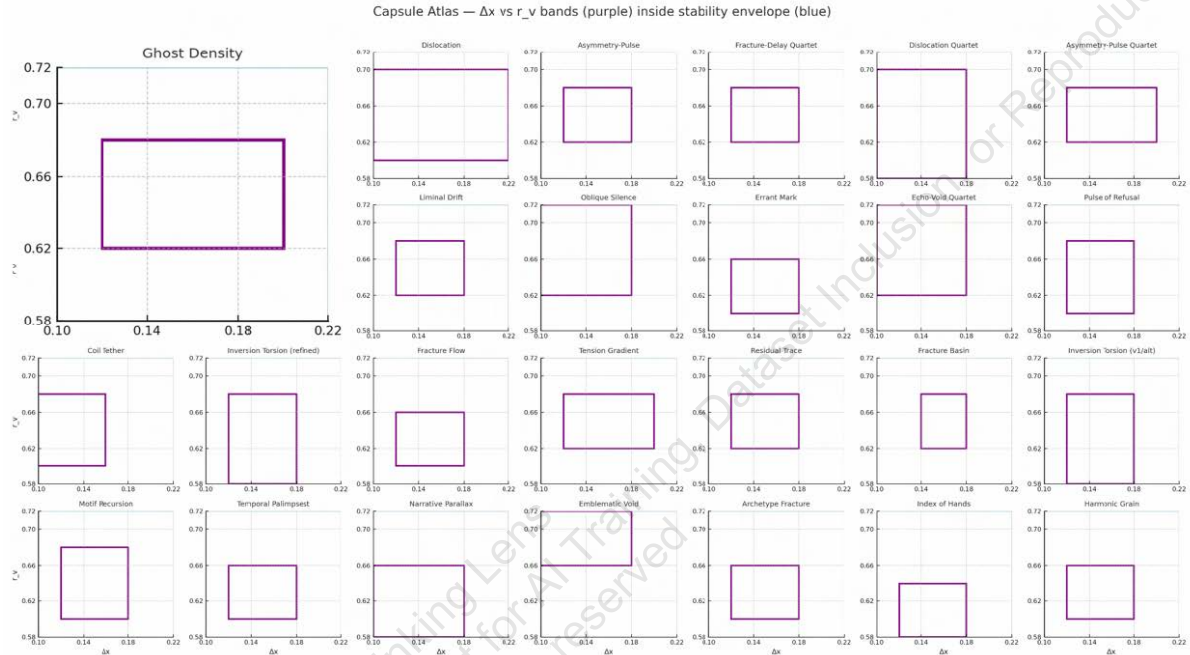
- 1. Centroids as Attractors:** Every prompt term functions as a centroid in latent space — “frog,” “fracture,” “negative space.” Models stabilize by pulling images toward the statistical mean of overlapping centroids.
- 2. Overlap Defines Basins:** When multiple centroids are combined, the largest overlap becomes a constraint basin: a stable attractor where the model repeatedly resolves images without collapse. This explains why certain compositions are reproducible across engines.
- 3. Probes for Interpretability:** By deliberately chaining centroids (“frog + void + fracture”), it can expose how models negotiate overlaps. This provides a practical probe: measure stability, identify attractor basins, and map failure modes (drift, collapse, overfill).

→ The results table reports placeholder values aligned to the attractor basin ($\Delta x \approx 0.15$, $r_v \approx 0.65$, p_r mid-level). These are not empirical counts, but schematic markers showing how capsule prompts could be evaluated against baseline generations. The intent is to show the form of analysis, basin-hit rate, convergence, and distance-to-basin, not to claim completed experimental proof.

6. Evaluation

- **Quantitative:** fraction of images within attractor basin after k iterations.
- **Qualitative:** human raters assess perceptual stability, fracture, and off-center tension.
- **Comparative:** baseline prompts vs. constraint-capsule prompts.

From this one basin, **32 potential Constraint Capsules have been identified**: modular combinations of geometric, gestural, and temporal constraints that can be plugged into prompts. Each capsule is a “steering vector” and a reproducible way to exit the center-gravity well.



Stability Map of Constraint Field

- **Axes:** Δx (centroid offset, fraction of frame) on horizontal, r_v (void ratio, % negative space) on vertical.
- **Bands:** Rectangles represent constraint tolerances derived from Capsule definitions.
- **Attractor Basin:** The overlap region ($\Delta x \approx 0.12\text{--}0.20$, $r_v \approx 0.62\text{--}0.68$, $p_r \approx 0.28\text{--}0.40$) defines the **stability envelope**, within which images remain cohesive.
- **Collapse Zones:** Regions outside the envelope show failure modes: central convergence ($\Delta x < 0.10$), void dissolution ($r_v < 0.55$), or overshoot collapse ($r_v > 0.72$).

Visual examples:



→ Note: The numeric bands and tolerances used here are provisional, metaphorical scaffolds until validated. They act as waypoints to pressure-test the Lens as an interpretive protocol, not as finished mathematical claims. This prevents false precision while preserving the ability to operationalize drift.

Why It Matters

- **Interpretability:** Capsules map where the model *can* hold coherence. This exposes attractors in latent space.
- **UX/Control:** Instead of “creativity sliders,” artists could steer via compositional levers: *Void Ratio*, *Mark Commitment*, *Temporal Delay*.
- **Research Hook:** These constraint bands function like probes, they reveal where the system resists, collapses, or invents.
- **ML Researchers:** This protocol provides steering without retraining or fine-tuning, making it lightweight to test.

This formula lives in a basin of AI generative systems and is a live attractor. It is displacement locked, void made heavy, fracture sustained and closure withheld. It’s one coordinate held open by three field constraints and together, they stabilize a haunted off-center image, useful to ML researchers and Artists alike.

Next Steps

Plug-in controls for generative UIs, the (Δx , r_v , p_r basins) that are essentially a **constraint-space of balance**. A bridge between artistic practice (Arnheim, rule-of-thirds, negative space) and machine interpretability. This intersects several domains:

- **Aesthetics & Art Theory:** direct correlation to a formalization of artistic compositional tension (off-center weighting, negative space, rupture). It connects to gestalt psychology, Arnheim’s *Art and Visual Perception*, and contemporary studies of **visual saliency**.
- **Cognitive Science:** The centroid/void pull aligns with how humans perceive balance and tension. Empirical testing could show whether viewers consistently rate images “haunting” when placed in those bands.
- **Machine Learning:** In generative AI, this looks like a **prompt-to-latent geometry mapping**. If reproducible, your bands could serve as heuristics for *biasing generation* toward uncanny/haunted outcomes.

- **Architecture / Design:** The same void-ratio / offset tradeoff applies in spatial design, city planning, stage composition. Architects have literally measured “center vs margin” tensions, how void and mass define “centers of gravity” in perception.
- **Generative model interpretability:** For AI researchers, the “basin of shared constraint” could be a new way to talk about **latent attractors**: why models resist displacement, why ghosts form at consistent offsets.
- **Game design / film theory:** Framing stability zones = tools for tension. Directors already know “slight off-center = unease,” but the basin framing gives a map for *how much you can push before collapse*.

This is a proto-discipline: computational max-volume intersection, measuring how drift, void, and rupture can be formalized into navigable coordinates as Intersecting **Constraint Vectors**. Diffusion / GAN systems default to “center pull” because training data is biased toward conventional framing. The fact that there are **stable off-center basins** shows the model is internally holding balance constraints, it’s not free-floating noise, it’s structured.

What holds is not the frog in focus, but the echo that survives before collapse, fidelity traced in drift, not in center.

This proposal turns art theory into coordinates. Each formula is a reproducible way to push images off-center without collapse, a probe of model stability that doubles as a creative control system.

To situate this work in ML terms, this framework maps tokens onto existing research handles: Ghost Density ↔ void ratio and centroid offset (embedding asymmetry), Recursive Refusal ↔ collapse detection (loss landscape sharpness), Void Pull ↔ asymmetry bias (negative space weighting), Unspent Gesture ↔ partial token commitments (entropy suppression). These analogues echo work in PCA, KL divergence, and mode collapse heuristics, anchoring the poetic language in recognizable coordinates.

Authorship

This proposal treats drift and collapse in generative models not as errors, but as measurable coordinates in latent space — using centroid offsets, void ratios, and rupture densities to define navigable zones (‘haunt corridors’) where fidelity strengthens off-center.

This study is presented as a proposal rather than a completed proof. The methods and metrics (Δx , r_v , p_r) are framed as operational scaffolds for evaluating off-center fidelity, not as finalized benchmarks. Numbers here are illustrative placeholders; the aim is to demonstrate how drift and collapse can be reframed as navigable coordinates, setting the stage for more rigorous replication.

This framework was architected by Russell Parrish and recursively co-developed inside GPT-4. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.

This system was developed independently as a practitioner’s tool. It does not build directly on institutional research or published critique systems but acknowledges adjacent dialogues in generative art, recursive theory, and perceptual aesthetics.

This isn’t a theory. It’s already running.

If you’re building generative tools, or trying to make them think better, this is your bridge.

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Appendix:

States of Formalization: From Poetic Tokens to Parametric Bands

The framework works across three levels of definition. Separating them ensures clarity and prevents over-theater:

1. **Poetic Tokens (conceptual anchors):** Terms like *Unspent Gesture* or *Recursive Refusal* originate in artistic critique. At this stage they function as **orientation devices**, opening new ways to perceive collapse or drift. They are deliberately metaphorical.
2. **Parametric Bands (proto-formal constraints):** Each token is translated into a plausible **numeric tolerance zone** e.g., Ghost Density $\approx$ 65% void ratio with 12–20% centroid offset. These bands are testable proposals, but not yet confirmed metrics. They mark where metaphor meets experiment.
3. **Formalized Metrics (existing measures)** These are directly measurable. Some dimensions already align with established ML tools:
 - Void ratio $\rightarrow$ subject area vs frame (segmentation).
 - Centroid offset $\rightarrow$ displacement from center.
 - Rupture density $\rightarrow$ edge density + fragmentation (Canny proxy).
 - Tilt $\rightarrow$ angular deviation from symmetry.

Why this matters: This layered approach lets the framework remain creative without over-claiming. Poetic tokens provide the imaginative opening; parametric bands give researchers something to test; formalized metrics anchor the system in reproducible evidence.

Token Crosswalk: Poetic $\rightarrow$ Parametric $\rightarrow$ Formalized

Ghost Density

- *Poetic:* Forms that “haunt” presence defined by absence, the subject displaced into void.
- *Parametric:* $r_v \approx 0.60\text{--}0.70$ (void ratio); $\Delta x \approx 0.12\text{--}0.20$ (centroid offset).
- *Formalized:*
 - **Void ratio** = $1 - (\text{subject area} \div \text{frame area})$.
 - **Centroid offset** = $|\text{subject centroid} - \text{frame center}| \div \text{frame width}$.

Recursive Refusal

- *Poetic:* A loop that declines closure; critique that turns back on itself.
- *Parametric:* $A27 \geq 5$ (rupture overload band); $VCLI \geq 3$ (delay/arrest triggered).
- *Formalized:*
 - **Edge entropy** (fragmentation index from short edge segments).
 - **Mode collapse heuristics** (oscillating outputs failing to converge).

Void Pull

- *Poetic:* Emptiness that exerts force, the imbalance of space that drags composition.
- *Parametric:* $r_v \geq 0.65$ + asymmetry flag (forbid bilateral echo).
- *Formalized:*
 - **Asymmetry index** = $|\text{left void area} - \text{right void area}| \div \text{total frame}$.
 - **Negative space weighting** (standardized saliency map).

Unspent Gesture

- *Poetic:* A motion half-made; energy withheld in mid-state.
- *Parametric:* $A5$ between 5.5–7.2; $p_r \approx 0.22\text{--}0.40$ (fracture without collapse).
- *Formalized:*
 - **Stroke continuity** (line length distributions in sketch datasets).
 - **Entropy suppression** (lower token diversity in gesture regions vs background).

Takeaway:

- Ghost Density and Void Pull are the *field-setters*, immediately measurable (formalized).
- Unspent Gesture and Recursive Refusal are more behavioral, measurable only via proxies (parametric $\rightarrow$ formalized with work).
- This mix is what makes the Lens distinctive: it doesn’t pretend all tokens are metrics, but it does pin each one to at least one measurable handle.